

Why spaCy for Resume Extraction

1. Why spaCy is Chosen

- Fast — can process hundreds of resumes quickly.
- Easy to customize — can be trained to find resume-specific fields like Name, Skills, Company, Degree.
- Good mix of rules + AI — regex catches email/phone, AI model catches names/companies/skills.
- Most existing resume parser tools already use spaCy as their base.

2. Accuracy Score (Compared to Other Tools)

Tool	Accuracy (Precision / Recall)
spaCy	0.96 / 0.97
NLTK	1.00 / 0.96 (misses some entity types completely)
Stanford CoreNLP	Slightly higher accuracy, but slow and hard to customize
BERT / Transformer models	Highest accuracy, but very slow, needs GPU

Conclusion: spaCy gives the best balance of accuracy, speed, and ease of customization for resume extraction.

3. Dummy Demo (Input to Output)

Input (Resume Text):

John Carter
Email: john.carter@email.com | Phone: +1-415-555-0192
Senior Data Analyst at Google, Mountain View
2019 - Present
Skills: Python, SQL, Machine Learning, Excel
Education: B.S. Computer Science, Stanford University, 2015

Output (Extracted by spaCy):

```
{
  "name": "John Carter",
  "email": "john.carter@email.com",
  "phone": "+1-415-555-0192",
  "organizations": ["Google", "Stanford University"],
  "dates": ["2019", "2015"],
  "locations": ["Mountain View"],
  "skills": ["Python", "SQL", "Machine Learning", "Excel"]
}
```